15 – Featurized MNL Choice Model

IEOR 145/245

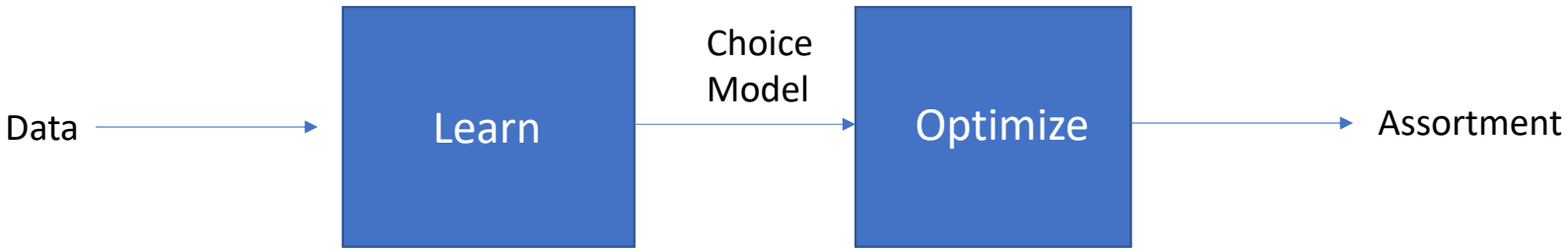
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Multinomial Logit Choice (MNL)

$$\phi(i,S) = \frac{v_i}{v_0 + \sum_{j \in S} v_j}$$

- Optimization and learning are tractable
- Often the go-to model, but has clear limitations

From Data to Decision



Sales Data

- Typical sample consists of the assortment shown and choice made

$$S_t, i_t\text{: customer } t \text{ saw assortment } S_t \text{ and chose } i_t \in S_t \cup \{0\}$$

- Available in brick and mortar stores as well as online platforms
  - Online platforms also have other data such as “clicks” or “hover”

- Dataset is a collection of **independent** samples $\{S_t, i_t\}_{t \in T}$
  - Called *sales data*

Learning MNL Model

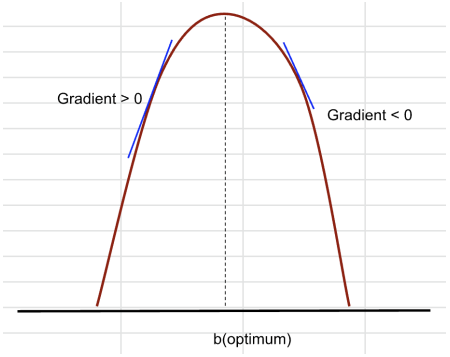
- Given sales data $\{S_t, i_t\}_{t \in T}$
- Log-likelihood estimate

$$= \text{argmax}_{(v_i)} \sum_t \log \frac{v_{i_t}}{1 + \sum_{i \in S_t} v_i}$$

($v_0 = 1$ as part of normalization)

$$= \text{argmax}_{(v_i)} \sum_t \log v_{i_t} - \sum_t \log \left(1 + \sum_{i \in S_t} v_i\right)$$

- Concave function, gradient ascent finds a solution efficiently



Important Question

What if data does not have item e, i.e., e is not shown in any of $\{S_t\}_{t \in T}$?

Log-likelihood estimate $= \text{argmax}_{(v_i)} \sum_t \log v_{i_t} - \sum_t \log\left(1 + \sum_{i \in S_t} v_i\right)$

Featurizing MNL

MNL is a Special Case of RUM

- Recall: RUM is defined by a joint distribution over utilities $(u_i)_{i \in I}$
  - Given assortment S, customer chooses item with highest utility
  - We can always decompose a random variables as
    μ_i $(deterministic\ mean) + \epsilon_i$ $(zero\ mean\ random\ part)$

- **MNL:** A RUM with $u_i = \mu_i + \epsilon_i \ \ \forall\, i \in [n]$ where
  - ϵ_i is **independently** distributed **Gumbel** random variable
  - μ_i is a constant (mean value of u_i)
  - In general RUM the ϵ_i can be correlated and need not be Gumbel

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Notice: where are the weights v_i?

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MNL Parameters from RUM

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  - μ_i is a constant (mean of u_i)
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- **Given the RUM definition of MNL, what is the choice probability**?

$$\phi(i, S) = Pr.\left(u_i > \max_{j \in S \cup \{0\}; j \neq i} u_j\right) = \frac{e^{\mu_i}}{\sum_{j \in S \cup \{0\}} e^{\mu_j}}$$

(derived from definition of Gumbel and independence of ϵ_i)

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$\Rightarrow$ MNL weights $v_i = e^{\mu_i} \;\; \forall i \in N$

How Does this Help?

- If weights come from utilities then ... we only need to learn utilities

- **Common hypothesis:** average utility is a function of item's features
  - Features capture a lot about usefulness of the item
  - For example, a few features such as color, brand, specs, price etc. may capture most important info

- We reduce all items (old and new) to their feature vector
  - Each dimension of the vector captures some aspect

Featured Mean Utility

$$\mu_i = \beta . X_i = \sum_k \beta_k \, x_{ik}$$

- β and X_i are **vectors**
- $X_i = (x_1, x_2, \dots)$ is a (known) vector of features of product i
  - Example features – price, quality, color, style etc.
- β_k is the importance/weight (could be negative) of k-th feature

- Inbuilt: items with similar features have similar utility
- Typically know the feature vector for all products
  - So if we find β, we can compute utility of every item

Featurized MNL Parameters

- Attraction parameters as a function of features:

$$v_i = e^{\beta . X_i}$$

Average utility of i, $\mu_i = \beta . X_i$

- Weight vector $\boldsymbol{\beta}$ **is the only unknown**

- Why is this useful?
  1. Once we learn β, we can compute v_i for any "new" product i!
  2. Unknown vector β is often low dimensional
     - 1 million products but β may just have 100 dimensions (100 unknowns)

Learning β from Sales Data

$$\log v_i = \beta . X_i \qquad (v_i = e^{\beta . X_i})$$

- Log-likelihood estimate

$$= \text{argmax}_{(v_i)} \sum_t \log v_{i_t} - \sum_t \log\left(1 + \sum_{i \in S_t} v_i\right)$$

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$$= \text{argmax}_{\beta} \sum_t \beta . X_{i_t} - \sum_t \log\left(1 + \sum_{i \in S_t} e^{\beta . X_i}\right)$$

- Concave in β
- Smaller dimensional problem
  1. Can learn β with fewer samples
  2. Don't need to have every item in the sales data!

Application to School Choice

[Not for Evaluation]

School Choice in US

- Many cities don't assign but give some *choices* to each family
  - Necessary for equal access to opportunities
  - Examples: Boston, Cambridge, Chicago, Denver, Minneapolis, Miami, NYC, New Orleans, San Francisco …

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  - Necessary for equal access to opportunities
  - Examples: Boston, Cambridge, Chicago, Denver, Minneapolis, Miami, NYC, New Orleans, San Francisco …

- **How should school system decide what choices to give?**

Tradeoffs in School Choice

- Why not give all possible options to each family?

Tradeoffs in School Choice

- Why not give all possible options to each family?
  - What if everyone wants to go to the same school?
    - Easy: use a lottery, which is fair and equitable
  - Any other reasons?

Tradeoffs in School Choice

- Why not give all possible options to each family?
  - What if everyone wants to go to the same school?
    - Easy: use a lottery, which is fair and equitable
  - **Unsustainably high transportation costs borne by the system**

- **Key tradeoff:** equal access *vs* geographic/budget limits

Case of Boston Public Schools (BPS)

Based on work by Peng Shi (USC):

https://pubsonline.informs.org/doi/full/10.1287/inte.2014.0781

The Overall Process

1. We give some options to each family

2. Families rank order their options

3. We assign schools based on family input

- We are interested in Step 1 – What options do we give each family?
  - Module 4 will touch on Step 3

The 3 Zone System of Choice

- A child can apply for any school in zone *or* within 1 mile from home
- Enacted in 1988
  - Zones decided based on socioeconomic balance
- How do you think it held up in 2012?



The 3 Zone System in 2003

- Times changed and so did socioeconomic balance

1. Quality of education in east zone was lowest
   - Imagine frustration of living on the edge of east-north zone!



The 3 Zone System in 2003

- Times changed and so did socioeconomic balance

1. Quality of education in east zone was lowest
2. High busing cost
3. Very unpredictable assignment
4. Community dispersion



From 2003 - 2012

- Several attempts at change

- **Main idea:** increase the number of zones (4, 5, 6, 12?)
  - Inequality persists, city councils couldn't agree

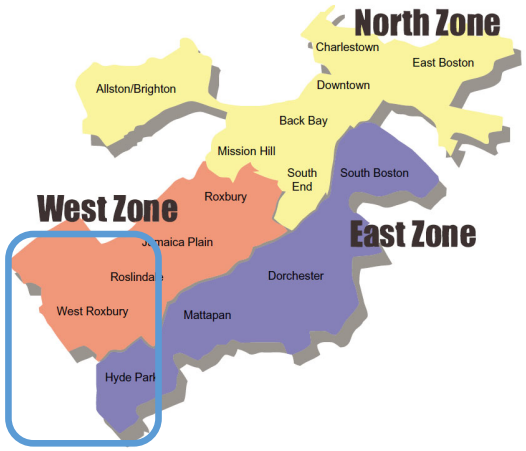
- 2012 – another almost failed attempt
  - 6-23 zone plan
  - Outside evaluation showed serious inequality in new plan
  - Operations Research PhD student to the rescue!

BPS Needed …

- A way to scientifically evaluate candidate policies
  - Without implementing each one

- **Given a policy** that determines the options for each family, **how do we predict the choices?**

How would people choose in this new zone?



Heart of the matter: Modeling Choice

- Given a set of options, student/family choice modeled using MNL choice model
- Population in each geocode modeled using separate choice model
- ϕ_t: characterizes how families in geocode t choose
- What if some schools were never an option in the past?
  - Use featurized choice model

MNL Model Estimation

- Consider families in a some geocode t
  - We want to learn ϕ_t

- **Data:** ranked preferences from past years
  - Sample 1: school 3 $>$ school 2 $>$ school 1
  - Sample 2: school 2 $>$ school 1 $>$ school 3
  - Sample 3: school 1 $>$ school 3 $>$ school 2
  - Sample 4: school 3 $>$ school 2 $>$ school 1
  - …

- Find MNL model that explains these preferences

Data

- In geocode t
  - Sample 1: school 3 $>$ school 2 $>$ school 1
  - Sample 2: school 2 $>$ school 1 $>$ school 3
  - Sample 3: school 1 $>$ school 3 $>$ school 2
  - Sample 4: school 3 $>$ school 2 $>$ school 1
  - …

- Is this related to sales data (S, i)?
  - Why care? – Because we know how to learn MNL from sales data

Data

- In geocode t
  - Sample 1: school 3 $>$ school 2 $>$ school 1
  - Sample 2: school 2 $>$ school 1 $>$ school 3
  - Sample 3: school 1 $>$ school 3 $>$ school 2
  - Sample 4: school 3 $>$ school 2 $>$ school 1
  - …

- Any 1 sample is equivalent to multiple sales data points
  - Sales sample 1: student sees $\{1,2,3\}$ and chooses 3
  - Sales sample 2: student sees $\{1,2\}$ and chooses 2

} school 3 $>$ school 2 $>$ school 1

Utilities

- Random student with geocode t
- j: school

$$u_{tj} = k_j - \beta_1 \text{Distance}_{tj} + \beta_2 \text{ Walk}_{tj} + \epsilon_{tj}$$

- k_j: perceived quality of school j (depends on various factors, unknown)
- Distance_{tj}: distance between geocode t and school
- Walk_{tj}: is the school in walking distance (1 mile)?
- ϵ_{tj}: unmeasured/unknown factors (Gumbel distribution)
- $k_j, \beta_1, \beta_2 \geq 0$: unknown

From Utilities to MNL Parameters

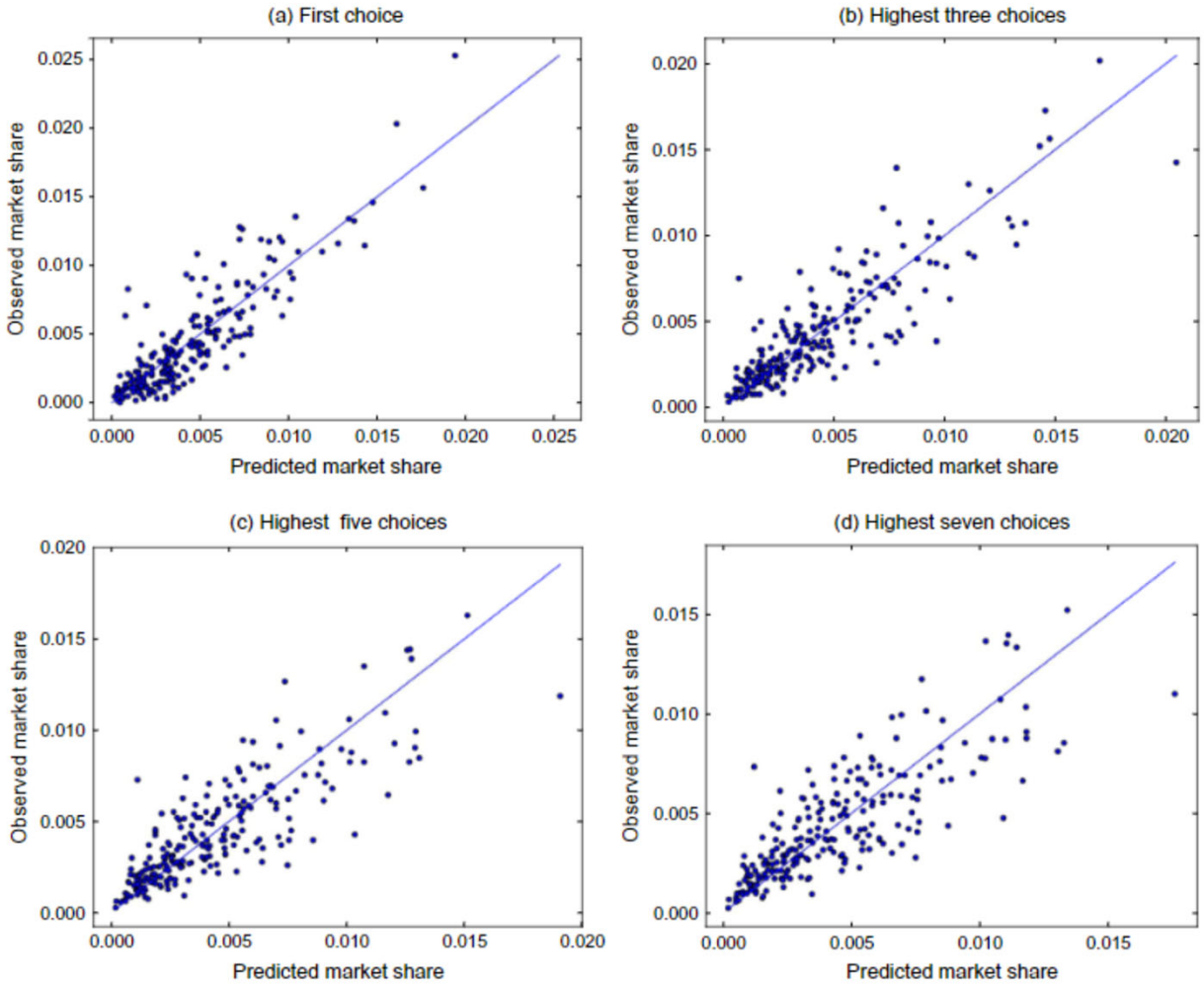
$$u_{tj} = k_j - \beta_1 \text{Distance}_{tj} + \beta_2 \text{ Walk}_{tj} + \epsilon_{tj}$$

- Probability that student in geocode/type t chooses school j given choices S:

$$\phi_t(j,S) = \frac{e^{(k_j - \beta_1 \text{Distance}_{tj} + \beta_2 \text{ Walk}_{tj})}}{\sum_{s \in S} e^{(k_s - \beta_1 \text{Distance}_{ts} + \beta_2 \text{ Walk}_{ts})}}$$

i.e., $v_{tj} = e^{(k_j - \beta_1 \text{Distance}_{tj} + \beta_2 \text{ Walk}_{tj})} \; \forall \; j \in S$

- No outside option (students need to go to some school)


(a) First choice
Observed market share
Predicted market share
0.000
0.005
0.010
0.015
0.020
0.025
(b) Highest three choices
Observed market share
Predicted market share
0.000
0.005
0.010
0.015
0.020
(c) Highest five choices
Observed market share
Predicted market share
0.000
0.005
0.010
0.015
0.020
(d) Highest seven choices
Observed market share
Predicted market share
0.000
0.005
0.010
0.015

Final Idea Adopted

- A family can list schools –
  - Within 1 mile of home
  - 2 closest schools ranked in top 25 % (according to reliable public metric)
  - 4 closest schools ranked in top 50 %
  - 6 closest in top 75 %
  - 3 closest “guarantee” schools

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1. Zone free (no discontinuities due to arbitrary boundaries)
2. Ages well with time as school ratings will incorporate changes

- Performance verified with simulation (using MNL choice model)
- After 10 years of failed attempts!

Exam

- This marks the end of the syllabus for the exam (everything till now)

- Will be in class on Oct 29 (80 minutes)
  - Last year's exam will be released soon
  - It was 60 mins, but you can expect similar difficulty (so you'll have more time)